جامعة أم القرى
UMM AL-QURA UNIVERSITY

CV Classification

Group 3
Section 2

ID	NAME
45505754	Rema Saed ALthqfi
445005178	Noran Abdullah Aljodi
445000219	Reema Salem Alharbi
445004475	Toleen Fadi Dahlawi
44510267	Ilaf Ahmed Bashawri

Dr. Afaf M. Almehmadi

Abstract:
This study explores how machine learning can be used to automate CV classification by analyzing unstructured resume text and predicting the most suitable job category for each applicant. The review highlights how traditional algorithms (Logistic Regression, Random Forest, SVM) and NLP techniques such as CountVectorizer and TF-IDF can convert raw CV text into meaningful numerical features that enable accurate classification. Prior research shows that manual resume screening often introduces human bias, inconsistencies, and long processing times, while machine learning models offer faster and more objective evaluation.

To assess model performance, the proposed system relies on traditional evaluation metrics including accuracy, precision, recall, and F1-score, providing a clear and standardized way to measure how effectively the models distinguish between job categories. In addition, the system incorporates fairness metrics to detect and mitigate potential bias in model predictions, ensuring more equitable and unbiased decision-making. By using these well-established performance metrics alongside fairness measures, the system ensures both transparent evaluation and responsible, interpretable model behavior.

The literature also identifies recurring limitations such as small datasets, lack of generalization, and bias in hiring data—highlighting the need for fairness-aware models and more diverse CV samples. By addressing these gaps, automated CV classification systems can enhance hiring efficiency, reduce bias, and support HR teams in making data-driven recruitment decisions.

Keywords: Machine Learning, CV Classification, NLP, Accuracy, Precision, Recall, F1-score, Hiring Automation, HR Tech

Introduction:
Resume classification is an increasingly important application of machine learning as companies face a growing number of applicants for each job position. The field focuses on using artificial intelligence to analyze unstructured resume text and automatically categorize applicants based on skills, experience, and qualifications. Instead of relying solely on manual reviewwhich can be slow, inconsistent, and influenced by human judgment machine learning models learn patterns from large sets of labeled CVs and use them to predict job categories for new applicants.

The system begins by extracting important information from resumes such as skills, education, experience years. These features are then transformed into numerical representations using NLP techniques, allowing classification models to detect patterns that distinguish each job category. This automated process helps recruiters identify suitable candidates more quickly and ensures a more objective and consistent evaluation process. By leveraging machine learning, organizations can reduce workload, improve fairness, and streamline the initial stages of candidate selection.

Problem Scenario
Companies receive large numbers of CVs for each position, making manual screening slow and inconsistent. Because resumes are unstructured text, extracting key skills and evaluating candidates objectively is difficult. Human judgment may also introduce unintentional bias, leading to unfair or inaccurate hiring decisions. These challenges limit the ability of HR teams to efficiently identify qualified applicants.

Objective of the Proposed System
The system aims to automatically analyze and classify CVs by extracting relevant skills and qualifications using machine learning and NLP techniques. Its goal is to support HR teams by reducing screening time, improving evaluation consistency, minimizing bias through fairness metrics, and enabling more accurate, data-driven hiring decisions.

Target Users
- HR professionals and recruitment specialists
- Companies with high-volume hiring
- Hiring managers
- Job platforms integrating automated CV screening

Challenges in the Current Problem
- Excessive manual work
- Human bias
- Difficulty extracting relevant skills from text
- Inconsistent evaluations
- Limited capacity for high applicant volumes

Proposed Machine Learning Solution
- Preprocess and clean CV text
- Convert text into numerical features using CountVectorizer
- Train classification models (Logistic Regression, Random Forest, SVM)
- Evaluate with accuracy, classification report, and confusion matrix
- Provide HR teams with ranked candidate lists
- This approach reduces bias, automates initial screening, and ensures faster, consistent decision-making.

Social, Academic, and Technological Relevance
- **Social:** Promotes fairness and objective candidate evaluation
- **Academic:** Applies NLP, machine learning, and AI ethics in bias mitigation
- **Technological:** Enhances HR Tech through intelligent CV classification

Conduct Literature Research

study (1), aims to address the fundamental problem of manual resume screening—which is often inherently time-consuming and labor-intensive—by implementing an automated e-recruitment platform. The primary objective was to enhance efficiency and minimize both human effort and errors, a necessity magnified by the widespread shift towards digital operations.

To tackle the challenge, the researchers in **(1)** employed a rigorous methodology rooted in Machine Learning and Natural Language Processing. After gathering unstructured data, a final clean dataset of **10,000 entries** was prepared, strategically split into **70%** for training and **30%** for testing across 25 distinct job categories.

The critical preprocessing steps included using **Lemmatization** (the process of reducing an inflected word to its dictionary root, ensuring higher linguistic accuracy) instead of **Stemming** (a simpler process that just chops off word endings without guaranteeing the resulting base has meaning).

The next crucial step involved using **TF-IDF Vectorization** (a technique that converts linguistic features into **quantifiable numerical vectors**, or numbers, which determine the relative importance of each skill by favoring words unique to a document).

Following data preparation, three supervised ML algorithms were deployed for comparative analysis, and the **detailed evaluation results** were as follows:

Algorithm	Accuracy (%)	Recall (%)	Precision (%)	F1 Score (%)
Naïve Bayes	**45**	44	46	45
Support Vector Machine (SVM)	**60**	58	61	60
Random Forest	**70**	69	71	70

Table1: Paper 1 Results

The results from **(1)** (As shown in Table 1) clearly indicated that the **Random Forest** algorithm was the superior performer, achieving the highest overall accuracy of **70%**, along with strong **Recall of 69%**, **Precision of 71%**, and an **F1 Score of 70%**. The model's efficacy was further validated by the Confusion Matrix.

However, the researchers acknowledged two significant constraints. Firstly, the 70% accuracy achieved was deemed **average** and not fully sufficient for critical recruitment decisions. Secondly, a critical **training efficiency trade-off** was identified:

attempts to further boost accuracy by increasing the number of decision trees (which were set at **2,500** trees) would inevitably lead to a **significantly slower learning time**, posing a major obstacle to the system's practical scalability. Considering these limitations, the proposed Future Work from **(1)** suggests ambitious extensions, including integrating the model with a frontend resume portal, and expanding the system's capabilities to incorporate advanced video interview features such as **facial recognition and voice analysis**.

As for Reference (2), we found that it addresses a strategic and critical challenge facing the talent acquisition function in Human Resources, which is the **arduous and time-consuming nature** of manually screening a huge volume of résumés, a problem compounded by the lack of a standard structure or format for these documents. Therefore, this study introduced a **two-phase automated model** based on Machine Learning, aiming to enhance recruitment efficiency and accelerate access to the most qualified candidates, thereby achieving significant savings in cost and time during the hiring process.

The researchers in **(2)** adopted a rigorous methodology that began with the **preprocessing** of the raw data.

This processing involved meticulous text cleansing, including the removal of special characters, numbers, and stop words, followed by the application of the **Lemmatization** technique to precisely extract the linguistic roots of the words.

After cleansing, the feature extraction phase utilized the **TF-IDF Vectorization** method, considering both **Uni-grams** and **Bi-grams**.

(This approach enabled the determination of the relative importance of terms and assigned higher weights to rare skills). To ensure the accuracy of generalization and the robustness of the results, the **10-fold cross-validation** technique was used throughout all stages of model evaluation.

In the first phase of the model, **Classification**, four traditional machine learning algorithms were tested to evaluate their efficiency in categorizing résumés into their job categories.

The test results showed a significant variance in performance; while algorithms such as Random Forest and Multinomial Naive Bayes recorded very low accuracy rates, the **Linear Support Vector Machine (Linear SVM)** classifier proved superior.

Classifier Evaluation Results:

Classifier	Abbreviation	Accuracy Score
Linear Support Vector Machine	Linear SVM	**0.7853** (78.53%)
Logistic Regression	Logistic Regression (LR)	0.6240 (62.40%)
Multinomial Naive Bayes	Multinomial Naive Bayes (NB)	0.4439 (44.39%)
Random Forest	Random Forest (RF)	0.3899 (38.99%)

Table 2: Paper 2 Results

Based on these results as shown (in Table 1) as the Linear SVM algorithm succeeded in finding the optimal separation level (**Hyperplane**) to distinguish data classes, it was adopted as the best model for the second phase, **Recommendation**. In this phase, the chosen classifier is used to match the categorized résumés against the provided Job Description, with the goal of providing the recruiter with a **ranked list** of candidates.

This ranking was achieved using two approaches: the content-based recommendation using the **Cosine Similarity** metric (It is a mathematical measure used to find the degree of content similarity between two items such as a résumé and a job description, **regardless of their size or length**. The focus is only on the **common direction (the angle)** between the mentioned words and skills; if the direction matches, then they are very similar.),

and the **k-Nearest Neighbors** algorithm to identify the closest candidates in the feature space.

In conclusion, the researchers in Reference **(2)** acknowledged that their model represents significant added value in first-level screening. However, they pointed out key challenges that must be addressed in future work, most notably:

- **File Format Constraints:** The current model requires data in **CSV format**, necessitating the integration of specialized text processing libraries (such as textract) to handle common **.pdf or .doc** files encountered in a real-world hiring environment.
- **Risk of Core Information Loss:** The researchers warned that the process of summarizing long texts using the gensim library might lead to the risk of **losing vital information** such as candidate skills and experience, demanding precise future tuning of the summarization process

study (2), proposes a system that uses Natural Language Processing and machine learning techniques to analyze resumes, extract skills, experience, and qualifications, and then compare them with job requirements to identify the most suitable candidates by generating a matching score that reflects each applicant's level of compatibility with the job. The system aims to speed up the screening process and reduce the effort required for manual resume evaluation.

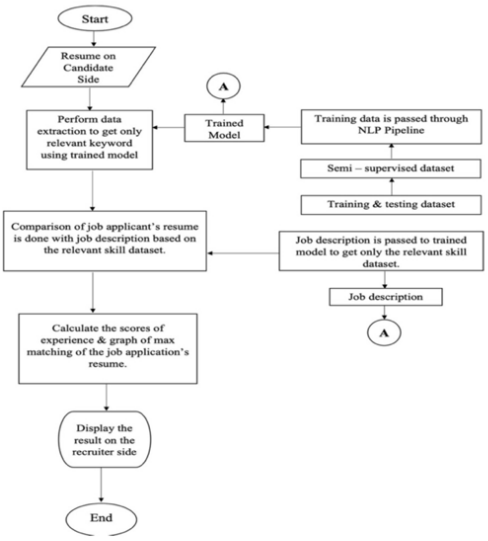

Figure1: Activity Diagram showing how the overall system works

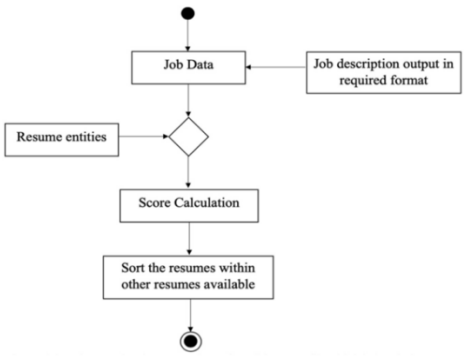

Figure2: Activity Diagram showing comparison of candidate's profile with job description

The system begins by receiving the resume in PDF format and converting it to plain text for easier digital processing. Next, it extracts keywords from the resume, including those from GitHub and LinkedIn profiles mentioned within the CV. Only the most relevant keywords are retained, while common, irrelevant ones are removed **(3)**.

The system also utilizes Named Entity Recognition (NER) technology within the spaCy framework to identify specific entities such as skills, experience, education, and certifications, enabling the extraction of accurate information from unstructured resumes **(3)**.

The study used a small dataset of 220 resumes **(200 for training and 20 for testing)**, along with job descriptions and skills manually annotated using **"Dataturks"**: (a tool typically used to prepare categorized data before training machine learning models). **The model underwent only initial training and testing trials, and no numerical performance metrics were reported**. Although the **researchers mentioned the use of the KNN algorithm during the data preparation phase**, the system does not rely on KNN as a classification model, nor does it incorporate any form of semi-supervised learning. Instead, the underlying model implemented in the system is spaCy Named Entity Recognition (NER), **as the project is still in its early stages of development(3).**

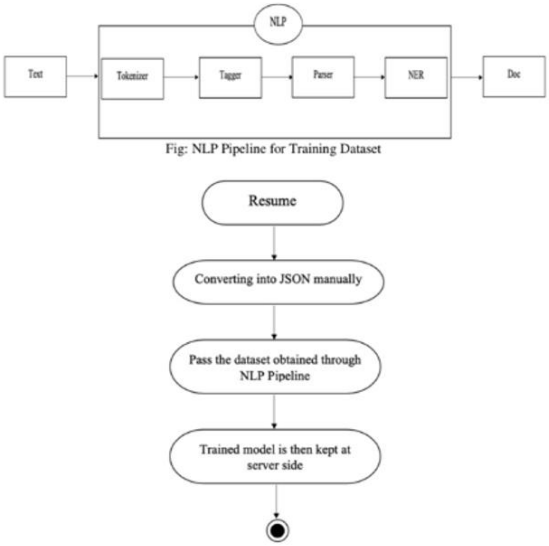

Figure 3 : Activity Diagram for training dataset

The system exhibits several limitations, the most significant of which include the small size of the dataset, the reliance on a simple keyword-based approach for skill extraction, difficulties in handling different resume formats, support limited to English-language resumes, and the unsuitability of the KNN algorithm for processing large or highly similar text datasets **(3)**.
Despite these limitations, the system is characterized by its simplicity and ease of implementation, providing a clear initial framework for automating resume evaluation. It also helps reduce manual effort and speeds up the screening process, making it a suitable foundation for developing more advanced systems in the future using deeper linguistic techniques and larger, more diverse datasets (**3**).

Research (4) proposes an integrated system for automating the process of classifying resumes into specific job categories. The researchers used four machine learning algorithms: **Logistic Regression, Random Forest, SVC, and KNN. Logistic Regression was chosen as the final model due to its simplicity, speed, and ease of deployment.**
The research relied on three ready-made datasets: **UpdatedResumeDataSet.csv, clean_resume_data.csv,** and **Resume.csv.** The data preparation phase included cleaning the text, removing common words, segmenting the words, and then converting the text into a numerical representation using **"TF-IDF"**: (a method of translating words in a text into numbers that express the importance of each word within the document compared to the rest of the documents). The data was then divided into **80% for training and 20% for testing**. The **"SMOTE"** technique was also used: (a smart method that increases the data of the lower category by creating new, artificial, but logical, samples to balance the data and allow the model to learn correctly) **(4)**.
Researchers used several evaluation metrics, including **accuracy, recall, positive accuracy, F1 score, Confusion matrix, and graphs**.
The results showed that logistic regression, random forests, and SVC achieved (99.48%) accuracy, while **KNN** achieved slightly lower **accuracy at (98.45%)**.
Based on these findings, a logistic regression model was implemented within a web application built using **"Flask"**: (a lightweight framework for building web applications using the Python language), allowing users to upload their resumes and receive predictions directly **(4).**

Performance Comparison,

Models	Accuracy
K-Neighbors Classifier (KNN)	0.9844559585492227%
Logistic Regression	0.9948186528497409%
Random Forest	0.9948186528497409%
Support Vector Classifier (SVC)	0.9948186528497409%

Table 3: Paper 4 Performance
Despite the system high performance (as shown in Table 3), it still suffers from some limitations, including its complete reliance on TF-IDF, which doesn't capture the deep meaning of the text, its lack of support for multiple languages, and its tendency to simply categorize resumes without matching them to specific job descriptions **(4)**.

Nevertheless, the system offers several important advantages, such as faster sorting, reduced human effort, improved recruitment efficiency, and ease of deployment and use **(4).**

Study(3), The research titled **ResuméAtlas: Revisiting Resume Classification with Large-Scale Datasets and Large Language Models** focuses on a resume classification system developed by researchers based on specific requirements, relying on data in this field.They collected **13,389 resumes** from Google Images, Bing Images, and LiveCareer , and then used **OCR techniques** for thorough data cleaning and text extraction. **(5)**

They used two types of models: **Traditional Models** and **Modern LLMs** (Large Language Models). Traditional models included: TF-IDF with SVM, XGBoost, Logistic Regression, Naive Bayes, MLP, and Random Forest. Modern LLMs included: **BERT** and **Gemma 1.1 2B**. The system's performance was evaluated using **Accuracy, Precision, Recall, F1-score, in addition to Top-k accuracy (As shown in Table 4). (5)**

Method Name	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
TF-IDF + Random Forest [33]	78.5	75.2	82.1	78.6
TF-IDF + SVM [1, 3, 6]	79.2	76.5	83.5	80.0
TF-IDF + Logistic Regression [3]	79.8	77.1	84.2	80.6
TF-IDF + Naive Bayes Multinomial [1, 3, 6]	81.3	79.5	86.2	82.8
TF-IDF + MLP [2]	81.6	80.2	87.1	83.6
BiLSTM [4]	81.8	78.3	83.5	79.8
TF-IDF + XGB [6]	83.5	82.1	89.5	85.8
BERT	91.2	90.5	92.8	91.6
BERT - Top 3	96.1	95.8	97.3	96.5
BERT - Top 5	97.5	97.2	98.1	97.6
BERT - Top 10	98.5	98.3	99.1	98.7
Gemma1.1 2B	**92.0**	**91.5**	**93.2**	**92.3**

Table 4: Paper 5 Performance

The system faced several problems, including: **variations in resume formats**, the **presence of sensitive personal data**, spelling errors, non-useful text (header, footer, links), heavy reliance on OCR accuracy, and multiple job titles for the same position. **(5)**

The system's drawbacks include: the need for high processing power, its sensitivity to errors resulting from OCR, weak interpretability of model decisions (Explainability), and difficulty in accurately handling resumes that contain more than one job. **(5)**

The system's advantages included: the development and creation of the largest Dataset in the field of AI-based resume classification, achieving high accuracy superior to previous research, using the latest Language Models (LLMs), providing open-source code, models, and Dataset, and a great diversity of jobs (43 categories) which improved its generalization ability. **(5)**

As for Reference (6), The research presents an automated resume screening system that uses Machine Learning models such as K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and One-vs-Rest. The dataset consists of a diverse collection of resumes that were preprocessed using NLP techniques like text cleaning, tokenization, POS tagging, and Named Entity Recognition to extract structured information from unstructured resume text. **(6)**

To evaluate the models, the authors used performance metrics such as accuracy. The results showed that KNN achieved the highest accuracy (98.44%), outperforming SVM and One-vs-Rest. This indicates that KNN was the most effective model for classifying resumes within the dataset used. **(6)**

Although the system performs well, the research highlights several challenges, such as inconsistent resume formats, possible dataset bias, and the need for more real-

world testing. The system's limitations include dependency on data quality and difficulty handling highly varied resume structures. However, its advantages are significant: faster screening, higher accuracy, reduced human bias, and improved fairness and efficiency compared to manual resume review. **(6)**

Model Used	Accuracy
KNN	98.44%
One v/s Rest	92.23%
SVM	94.76%

Table 5: Paper 6 Performance

Study 4

In paper 7, it addresses the challenge of manually screening resumes, which is slow, tedious, and prone to human bias. The authors propose an automated resume classifier that uses NLP to convert resumes into numerical vectors and applies machine learning to categorize them by job type. By focusing on skills and experience rather than personal identifiers such as name or gender, the approach has the potential to reduce human bias during the initial screening [Reference 7].

They used a Kaggle dataset containing 962 resumes across 25 job categories, including Data Science, Arts, Web Design, Sales, Health & Fitness, Business Analysis, and Electrical Engineering. This constitutes a multi-class classification problem rather than a simple good-versus-bad evaluation. [7]

The methodology includes preprocessing steps such as tokenization, stopword removal, lemmatization, and creation of frequent n-grams to preserve multi-word skills like "machine learning." For representation, each word in the resumes was converted into a 20-dimensional vector by training a Word2Vec skip-gram model on the corpus using Gensim. Each resume was represented by the sum of its word embeddings, resulting in a single 20-dimensional vector. [7]

For classification, five algorithms—SVC, Decision Tree, Random Forest, KNN, and Naive Bayes—were evaluated using 10-fold cross-validation on 80% of the dataset. The model with the highest weighted F1-score was selected and tested on the remaining 20% of the data, evaluated using precision, recall, and weighted F1. **Decision Tree** slightly outperformed the others, achieving weighted precision = 1.0, recall = 0.99, and F1 = 1.0 on the test set, indicating near-perfect performance. [7]

The study's strengths include a clear pipeline, rigorous evaluation metrics beyond accuracy, and systematic model comparison. Limitations include the small dataset, lack of algorithmic bias evaluation, and potential overfitting of the Decision Tree model, which may explain the extremely high test performance. [7]

In paper 8, it combines a survey with a practical framework, focusing on two interconnected tasks: shortlisting resumes and classifying them into job categories.

Manual resume screening is slow, biased, and unscalable, especially for large companies. The authors propose using NLP and machine learning to automate both tasks, improving efficiency, and reducing bias [Reference 8].

They used a Kaggle IT resume dataset with 963 resumes across 25 categories, similar to Paper 7. Preprocessing involves removing irrelevant features such as email and phone numbers, handling missing values, and performing lemmatization. Text was converted into numerical vectors using Bag-of-Words or TF-IDF, with TF-IDF emphasizing rare but important words. [8]

For shortlisting, resumes and job descriptions were represented as vectors, and cosine similarity was used to rank candidates by relevance. Other similarity measures, including Jaccard and Euclidean distance, were considered, but cosine similarity was the main approach. For classification, KNN, SVM, and Random Forest were evaluated to categorize resumes into 25 job classes. [8]

Results showed that **SVM** achieved the highest accuracy (99.48%), and cosine similarity effectively ranked resumes according to job descriptions. The study's strengths include a clear separation between shortlisting and classification, comparison of multiple algorithms, recognition of human bias, and a comprehensive review of related work. [8]

Limitations are close to Paper 7: the dataset is relatively small, evaluation relies heavily on accuracy without examining fairness or robustness, and algorithmic bias is not measured, despite acknowledging human bias in manual screening. [8]

Study (5), Paper 9: Resume Classification System using Natural Language Processing and Machine Learning Techniques (2022) This paper addresses automating the recruitment process by using Natural Language Processing (NLP) and Machine Learning to classify job applications. The objective was to develop an automated Resume Classification System that compares various ML algorithms to accurately classify resumes into 25 distinct job domains such as Java Developer, HR, and Testing.
The researchers used a dataset of 962 parsed resumes in CSV format. The preprocessing pipeline included data cleaning (removing URLs, special characters, and stop words using NLTK), Lemmatization to reduce words to their root form, and TF-IDF Vectorization with 1500 max features to convert text into numerical vectors. They trained nine classifiers using a 70-30 train-test split:
Support Vector Machines (Linear SVC, SGD, NuSVC), Naïve Bayes variants (Bernoulli, Multinomial, Gaussian), KNN, and Logistic Regression.
The Linear Support Vector Classifier (LSVC) achieved the highest performance with 99.6% accuracy and near-perfect precision/recall of 1.00, indicating the data was linearly separable. Logistic Regression followed with 99.3% accuracy, Multinomial Naive Bayes with 94.8%, while Bernoulli Naïve Bayes performed worst at 79.2%.
The study's main strength lies in its comprehensive comparison of nine algorithms with clear performance visualizations. However, significant limitations exist: the dataset of only 962 resumes is small for generalization, substantial class imbalance

exists (Java Developer had 81 resumes while Advocate had only 20), and the near-perfect accuracy suggests potential overfitting that may not translate to real-world performance.
For our project, this study demonstrates SVM's power but highlights a critical gap: the small, imbalanced dataset. While SVM works well with TF-IDF features for keyword-based classification, modern recruitment systems need models like BERT or LSTM that capture deeper semantic meaning beyond simple keyword counting, justifying our exploration of Deep Learning approaches with larger, more balanced datasets.

Paper 10: CVs Classification Using Neural Network Approaches Combined with BERT and Gensim: CVs of Moroccan Engineering Students (2024)
This study explores CV classification using Deep Learning architectures combined with advanced word embedding techniques. The research focuses on comparing neural network models (GRU, LSTM, CNN) and their hybrid combinations when paired with BERT and Gensim Word2Vec embeddings to classify Moroccan engineering students" CVs into five academic specialties.
The researchers collected 867 CVs from ENSAK engineering students representing five different specialties. The raw data came in diverse formats including Docx, PDF, PNG, and JPEG files. The preprocessing pipeline involved extracting text from these various formats into a unified CSV file, followed by data cleaning that removed special characters, punctuation marks, and numbers. For feature representation, they compared BERT embeddings against Gensim Word2Vec, then trained individual neural network models (GRU, LSTM, CNN) as well as hybrid architectures (CNN-GRU, LSTM-CNN) using an 80-20 train-test split.
The results showed that BERT-based models consistently outperformed Gensim Word2Vec across all architectures. The hybrid CNN-GRU model with BERT achieved the highest accuracy of 93.51% with precision of 0.9310 and recall of 0.9411. Single-architecture CNN with BERT reached 91.88% accuracy, while the LSTM-CNN hybrid using Gensim achieved 90.25%. Hybrid models generally performed better than single architectures by leveraging both sequential and convolutional features.
However, the study has notable limitations. The dataset of 867 CVs is relatively small for training complex Deep Learning models, and the focus on Moroccan engineering students limits generalizability to other regions or professional domains. For our project, this research demonstrates the power of Deep Learning and BERT embeddings for CV classification but also highlights the persistent challenge of dataset size and the need for larger, more diverse training data to build truly generalizable systems.

Paper Title and Release Year	Problem	Proposed ML-Based Solution	Dataset Description	Performance Results	Advantages	Weaknesses
2022 Resume Classification using various Machine Learning Algorithms	**Arduous manual screening**; need to **automate** the first stage of the hiring process (Classification).	**-Naïve Bayes** **-SVM** **-Random Forest**.	datasets from various websites like kaggle.com, glassdoor.com and indeed.com.	Naïve Bayes **=45%** Support Vector Machine (SVM) **=60%** Random Forest **=69%**	-Enhance efficiency -reduces human error -- uses **linguistically accurate Lemmatization**.	**-70% accuracy** deemed *average*. **Speed/Accuracy Trade-off** when increasing the number of Decision Trees.
2020 A Machine Learning approach for automation of Resume Recommendation	**Arduous manual screening**; lack of standard CV format; difficulty mapping CV content to Job Description (JD).	-Linear SVM -Logistic Regression -Multinomial Naive Bayes -Random Forest	CVs/JDs from online portals/Kaggle	Linear SVM = **78.53%** Logistic Regression = **62.40%** Multinomial Naive Bayes = **44.39%** Random Forest **=38.99%**	Full automation of initial screening; provides CV **ranking** based on JD fit; leads to cost/time savings.	-Input restricted to **CSV format.** -Risk of **information loss** during text summarization (gensim library).
2022 CV Analysis Using Machine Learning	The large number of resumes makes selection difficult, and current systems only use keyword matching without accurate ranking.	spaCy NER. Dataturks annotation. NLP pipeline.	- 220 resumes: 200 training 20 testing -Job descriptions. -Manually annotated skills using Dataturks.	The system is still under development. Only initial training/testing was performed. No numerical performance metrics were reported.	Automate the resume screening process. Extract essential information using NLP. Reduce the workload for recruiters by ranking candidates based on their	Limited data. only English support. simple skills extraction.

					suitability for the job.	
2025 Resume Classification using Machine Learning	Manual resume screening is slow and inefficient.	TF-IDF for text features. Logistic Regression. Support Vector Classifier (SVC). Random Forest. K-Nearest Neighbors (KNN).	UpdatedResumeDataSet.csv. clean_resume_data.csv. Resume.csv.	Best accuracy: **99.48%** (LR, SVC,RF) KNN: **98.44%**	High accuracy. fast screening. simple deployment via Flask. reduces HR workload.	Relies only on TF-IDF. no deep learning. no multilingual support. only classification (not job-matching).
2024 ResuméAtlas: Revisiting Resume Classification with Large-Scale Datasets and Large Language Models	Existing resume classification models suffer from low accuracy due to small datasets, non-standard resume formats, noisy text, and privacy-related limitations in collecting high-quality training data.	The study applies traditional ML methods (TF-IDF with SVM, XGBoost, Logistic Regression, Naive Bayes, MLP, Random Forest) and advanced LLM-based models (BERT and Gemma 1.1 2B) to build	ResuméAtlas dataset consisting of 13,389 resumes across 43 job categories, collected from Google Images, Bing Images, and LiveCareer, with text extracted using OCR and heavily preprocessed.	• BERT: 91% Top-1 accuracy, 97.5% Top-5 accuracy • Gemma 1.1 2B: 92% accuracy • Best classical model (XGBoost): 83 .5% accuracy • Evaluation metrics: Accuracy, Precision, Recall, F1-score, Top-k accuracy	- Largest resume classification dataset to date (13k+ samples, 43 labels) - High accuracy achieved using advanced LLMs (BERT and Gemma) - Models, code, and dataset are open-source and reproducible - Wide variety of job categories	• Heavy dependence on OCR accuracy; text extraction errors affect performance • High computational cost; requires GPU resources for training • Limited explainability of LLM decisions • Resume text is noisy and highly variable, which still poses challenges • Multi-label classification is

		a more accurate and robust resume classificatio n system.			improves generalization - Strong performance improvement over traditional ML approaches	not fully solved (handled only via Top-k predictions)
2024 Resume Screening Using Machine Learning	Manual resume screening is slow, inconsistent, and biased. Recruiters struggle with large volumes of resumes and often miss qualified candidates.	The study uses NLP for text extraction and applies ML models (KNN, SVM, One-vs-Rest) to automaticall y classify resumes based on skills and experience.	A diverse set of resumes processed with cleaning, tokenization, normalization, and labeled for skills and experience.	• KNN: 98.44% (best) • SVM: 94.76% • One-vs-Rest: 92.23%	High accuracy, reduced human bias, fast processing, scalable, and effective resume information extraction with NLP.	Varied resume formats, possible dataset bias, not tested in real workplaces, and limited by traditional ML models.
Automating Resume Classificati on: Leveraging NLP and AI for Efficient Candidate Screening (2023)	Manual resume screening can be inconsistent and slow	- Text preprocessi ng like Word2Vec embeddings -ML classifiers: Decision Tree Random Forest SVM & KNN Naive Bayes	Kaggle resume dataset with: 962 resumes From 25 different job categories	-Very high performance -Decision Tree had best results around F1 ≈ 0.98 Precision ≈ 1.0 Recall ≈ 0.99 - They used cross validation of 10 folds	-clear NLP pipeline -tried many ML models -detailed evaluation -almost perfect results	- very small dataset - almost no bias or fairness discussion - extremely high accuracy = maybe overfitting

Machine learning based Resume Shortlisting and Classification (2024)	Companies receive thousands of resumes and HR can't review them fast enough, so shortlisting becomes slow and inconsistent.	-shortlisting using text similarity - classification using ML models such as: SVM , KNN Random Forest	Kaggle dataset again about 963 resumes around 25 different job fields	- SVM performed best with accuracy of 99.48% - The accuracy of KNN and Random Forest were very high too, 99-98	-realistic idea -practical steps and clear text preprocessing -strong performance	-very small dataset - almost no fairness or bias evaluation -evaluation only focused on accuracy
Resume Classification System using Natural Language Processing and Machine Learning Techniques (2022)	Manual resume screening is time-consuming and inefficient; need to automatically classify resumes into specific job categories	Nine classifiers compared: Linear SVC, SGD, NuSVC, Bernoulli NB, Multinomial NB, Gaussian NB, Complement NB, KNN, Logistic Regression + TF-IDF (1500 features) + NLTK preprocessing & lemmatization	962 parsed resumes in CSV format (25 job categories, e;g; Java Developer, HR, Advocate, Testing, etc.)	Linear SVC: **99.6%** accuracy (precision/recall ≈1.00) Logistic Regression: 99.3% Multinomial NB: 94.8% Bernoulli NB: 79.2% (worst)	Comprehensive comparison of 9 algorithms with clear visualizations; Linear SVC extremely effective with TF-IDF on this task	Very small dataset (962 resumes), severe class imbalance (81 Java Developer vs 20 Advocate), near-perfect accuracy strongly suggests overfitting; results unlikely to generalize in real-world scenarios
CVs Classification Using Neural Network	Need for accurate automated classification of student	Deep Learning models: GRU, LSTM,	867 CVs of Moroccan engineering students (ENSAK) in	Best: **CNN-GRU + BERT → 93.51%** accuracy (precision	BERT embeddings significantly outperform Word2Vec;	Relatively small dataset (867 CVs) for deep learning, limited to Moroccan

Approaches Combined with BERT and Gensim: CVs of Moroccan Engineering Students (2024)	CVs into academic specialties; traditional methods lack semantic understanding	CNN and hybrids (CNN-GRU, LSTM-CNN) Embeddings compared: BERT vs Gensim Word2Vec	Docx/PDF/PNG/JPEG → converted to CSV (5 academic specialties) 80-20 train-test split	0.9310, recall 0.9411) CNN + BERT: 91.88% All BERT-based models > Gensim Word2Vec models Hybrids > single architectures	hybrid CNN-GRU architecture achieves strong results; demonstrates value of deep semantic understanding	engineering students (poor generalizability to other countries or professional domains), no comparison with traditional ML baselines

Table 6: Comparing Between 10 Papers

Literature Gaps :
Following the review of existing literature, two significant research gaps were identified, both of which have not been sufficiently explored in previous studies…

The first gap is related to how the text is represented. Most of the studies relied on TF-IDF as the main method for processing resumes. Although TF-IDF helps identify important words, it does not understand the actual meaning or context of sentences within the resume. Instead, it focuses only on word frequency, which makes the model sensitive to differences in writing style rather than the real skills and experience being described. As a result, this limits the ability of previous models to generalize or handle different ways of expressing the same skill.

The second gap is the lack of a clear evaluation of bias in the models used. Most of the research focused on traditional performance metrics such as accuracy, recall, positive accuracy, F1 score, and the confusion matrix, but did not examine whether the models might make unfair decisions based on writing style, language, or other factors. Since resume classification is directly linked to the hiring process, and recruitment is a sensitive field that requires fairness and transparency, ignoring bias is considered a major limitation in earlier research.

Our Dataset

The dataset consists of a collection of resumes labeled with various job categories. Each entry includes free-text resume content, candidate experience information, and a target job category. The dataset is multi-class in nature and exhibits noticeable imbalance across categories, with some job roles being heavily represented while others contain very few samples. As shown in Figure 4, this imbalance affects both model training and

fairness evaluation, and therefore required additional preprocessing steps such as data cleaning, label normalization, and oversampling during model development.

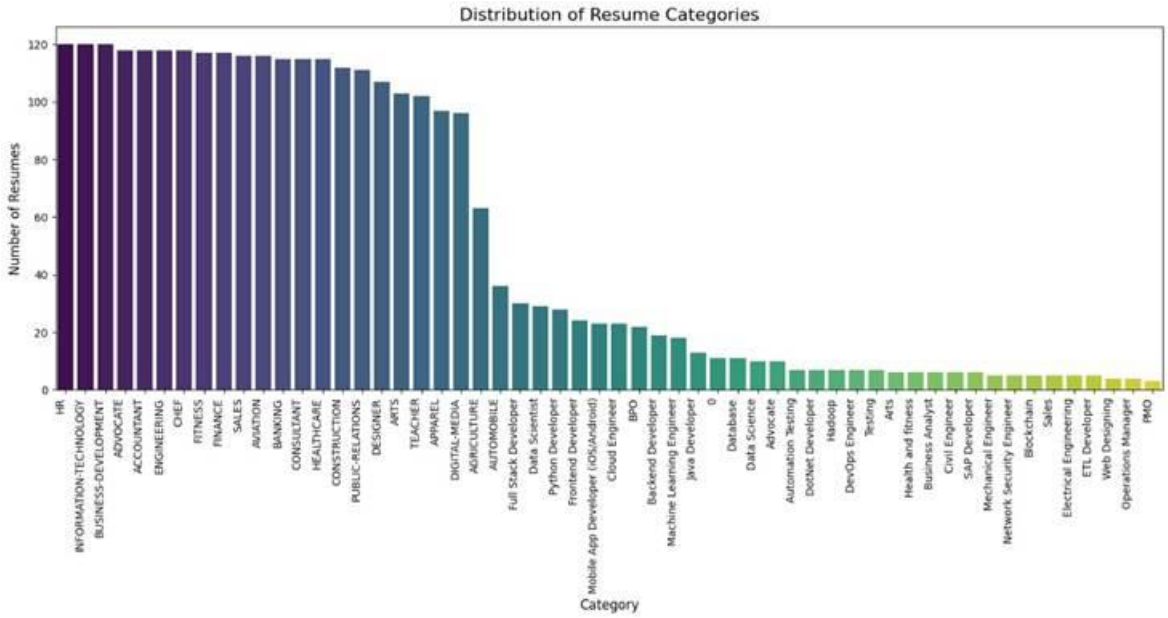

Figure 4: Distribution of Resume Categories *(Shows category imbalance and label distribution.)*

Detailed Machine Learning Module

This section presents the detailed design of our proposed solution. It outlines the model selection, the methodological approach, the workflow, and how this module integrates into a real-world recruitment ecosystem.

1. Objectives

The machine learning module in this project aims to achieve two main goals:

1. **Resume Category Classification (Problem 1)**
   Automatically classify resumes into predefined job categories using semantic text representations, enabling faster screening and improved HR processing efficiency.

2. **Fairness and Bias Assessment Across Experience Levels (Problem 2)**
   Evaluate whether classification decisions disproportionately favor or disadvantage specific experience groups (junior, mid, senior) and apply fairness constraints to reduce unjustified disparities.

These two objectives together create a system that is both **high-performing** and **responsible**, ensuring that automated decisions remain transparent and equitable.

2. Proposed Approach

To meet the objectives above, the ML solution uses a two-pipeline strategy:

Pipeline A – Resume Classification (SBERT + Logistic Regression + SMOTE)

- Use Sentence-BERT (*all-MiniLM-L6-v2*) to extract 384-dimensional semantic embeddings.
- Apply preprocessing and SMOTE oversampling to address class imbalance.
- Train a Logistic Regression classifier with class balancing.
- Evaluate using precision, recall, F1, and ROC-AUC.

Pipeline B – Fairness Assessment (TF-IDF + SVM + Fairlearn)

- Represent text using TF-IDF with unigrams and bigrams.
- Train a baseline Linear SVM classifier.
- Compute fairness metrics: demographic parity difference/ratio, equalized odds difference, and group-wise metrics per experience level.
- Apply fairness constraints using ExponentiatedGradient with Equalized Odds.
- Compare baseline vs fairness-aware models visually and numerically.

This dual approach allows the system to deliver **high-accuracy classification** while also **quantifying and correcting model bias**.

3. Workflow Diagram

The full machine learning workflow is shown in **Figure 5**.
It integrates both pipelines into a single processing path:

1. Load dataset
2. Clean the dataset and preprocess the text
3. Encode and prepare labels
4. Split sets into train and test sets
5. Branch into:
    a. SBERT → SMOTE → Classification
    b. TF-IDF → SVM → Fairness assessment

6. Output predictions, fairness metrics, and comparison visualization.

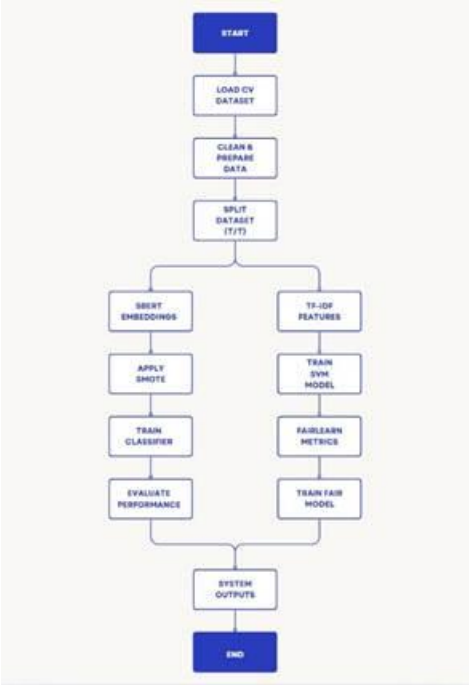

Figure 5: Unified ML Workflow for Resume Classification and Fairness Analysis

4. Integration into the Larger System

The ML module is designed to operate as a component of a broader resume-processing platform. It integrates in the following ways:

- **Classification API Endpoint**

  The SBERT model can be deployed as a service using:

  POST /predict_category

  ```
  {
      "resume_text": "...",
      "skills": "..."
  }
  ```

The endpoint returns the predicted job category along with optional confidence scores.

- **Fairness Monitoring Component**
  The fairness pipeline does **not** interfere with real-time predictions.
  Instead, it runs as an **audit module** to ensure responsible AI use by detecting disproportionate outcomes across experience groups.
- **Modular Architecture**
  Each component (preprocessing, embedding, classifier, fairness evaluator) can be independently upgraded or replaced.

5. Expected Benefits

The system contains many advantages:

Performance Benefits

- High-quality semantic embeddings produce significantly stronger classification performance compared to TF-IDF alone.
- SMOTE and class-weight balancing improve generalization for minority categories.
- ROC-AUC scores above 0.97 indicate excellent class separability.

Fairness & Responsible AI Benefits

- Identifies hidden disparities between junior/mid/senior applicants.
- Provides measurable fairness indicators (EOD, DP difference, group-wise F1).
- Applies corrections that significantly reduce unfairness in well-represented categories.

System-Level Benefits

- Scalable and deployable as an API.
- Easy to maintain and extend for additional bias dimensions (gender, nationality, etc.).
- Provides HR stakeholders with transparent and accountable automated screening tools.

6. Methodology

6.1 Data Loading and Initial Exploration

The dataset was inspected to evaluate completeness and structure. Several rows contained missing values in critical fields, particularly text fields. These rows were **removed rather than imputed**, because imputing text would introduce synthetic content with unknown semantic meaning. Only a small percentage of rows were affected, ensuring dataset integrity.

Summary statistics were generated to examine numerical stability and detect outliers.

6.2 Data Cleaning

Cleaning steps were designed to remove corrupted data while preserving meaningful content.

1. **Removing missing data:**
   Text fields cannot be realistically imputed, and missing labels compromise supervised learning. Dropping such rows prevents the model from learning from artificial placeholders.
2. **Filtering unrealistic experience values:**
   Experience was restricted to 0–50 years. Values above 50 likely arise from data-entry errors, while 50 remains within realistic interpretability.
3. **Text preprocessing:**
   a. Lowercasing
   b. Removing punctuation
   c. Lemmatization (chosen over stemming to preserve word semantics)

Although punctuation removal may reduce information in terms like “C++,” semantic embedding with SBERT mitigates this limitation.

6.3 Category Balance Assessment

As shown in **Figure 4**, the dataset exhibits significant class imbalance. Several categories have very small sample sizes, which negatively affects classifier performance and fairness calculations. To address this:

- A **stratified split** preserves category proportions.
- **SMOTE oversampling** is *performed* ***only on the training set*** *to prevent data leakage.*
- **Class weights** help correct imbalance during model training.

Inconsistent category labels (e.g., "Arts" vs "ARTS") introduce label noise, which inherently limits maximum accuracy.

6.4 SBERT Representation

The Sentence-BERT model (*all-MiniLM-L6-v2*) was used to produce 384-dimensional embeddings. SBERT captures semantic similarity and is highly effective for resume text.

Embedding produced a matrix of shape **(2770 × 384)**, which served as input for classification.

6.5 Classification Model

The SBERT embeddings were used in a Logistic Regression classifier trained with:

- A train–test split that preserves class proportion
- SMOTE oversampling
- Class-weight balancing

Results **(Figures 6-7)**:

- **Accuracy ≈ 0.73**
- **Macro F1-score ≈ 0.66**
- **ROC-AUC ≈ 0.97**

```
Accuracy : 0.7292418772563177
Precision: 0.6701677786656749
Recall   : 0.6813455485641883
F1-Score : 0.6662827262613288
ROC-AUC  : 0.9716650726782862
```

Figure 6: SBERT Classifier Performance Metrics

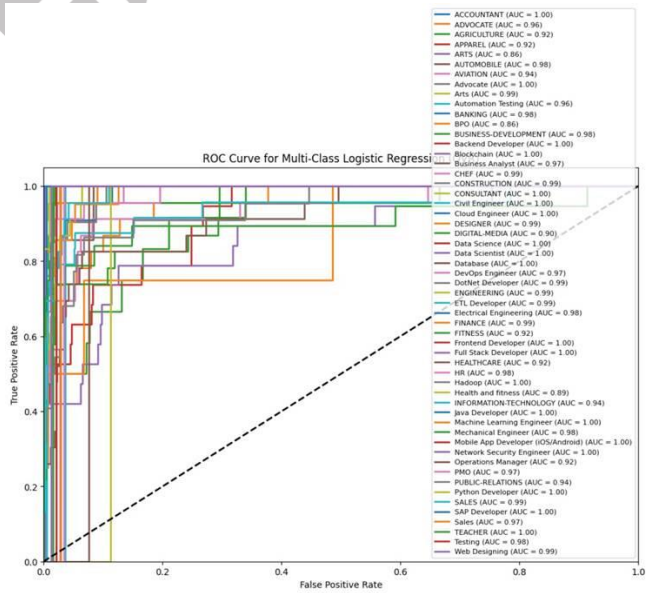

Figure 7: ROC Curves for Multi-Class Logistic Regression

6.6 Fairness & Bias Assessment

The fairness pipeline uses TF-IDF + SVM. Fairness was tested across experience groups using:

- Demographic Parity Difference
- Demographic Parity Ratio
- Equalized Odds Difference
- Group-wise precision, recall, F1

A fairness-aware model using ExponentiatedGradient enforced Equalized Odds constraints.

Results for HR appear in **Figures 8–10**.

```
--- Baseline SVM Performance ---
Accuracy:  0.9856
Precision: 0.8077
Recall:    0.8750
F1-score:  0.8400

--- Baseline SVM fairness ---
Demographic Parity Difference : 0.0110
Demographic Parity Ratio      : 0.8011
Equalized Odds Difference     : 0.1667

Group-wise metrics (SVM):
           accuracy    recall  precision        f1
exp_group
junior     0.987013  0.833333   0.882353  0.857143
mid        1.000000  1.000000   1.000000  1.000000
senior     0.976378  1.000000   0.571429  0.727273
```

Figure 8: Baseline SVM Performance and Fairness Metrics (HR)

```
--- Fair Model (EG + LR) performance ---
Accuracy:  0.9856
Precision: 0.7667
Recall:    0.9583
F1-score:  0.8519

--- Fair Model (EG + LR) Fairness ---
Demographic Parity Difference : 0.0097
Demographic Parity Ratio      : 0.8247
Equalized Odds Difference     : 0.0556

Group-wise metrics (EG + LR):
           accuracy    recall  precision        f1
exp_group
junior     0.987013  0.944444   0.809524  0.871795
mid        1.000000  1.000000   1.000000  1.000000
senior     0.976378  1.000000   0.571429  0.727273
```

Figure 9: Fair Model Performance Using Exponentiated Gradient (HR)

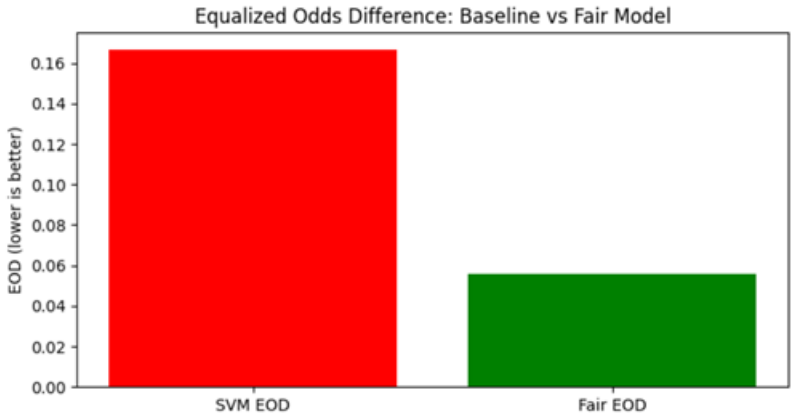

Figure 10: Equalized Odds Difference Before and After Fairness (HR)

6.7 Fairness Summary Across Categories

Fairness was tested on five categories: HR, Accountant, Business-Development, IT, Fitness. Results show (Table 7):

- HR & Accountant: strong improvement
- Fitness: minor improvement
- IT: no improvement
- Business-Development: fairness worsened due to extremely small data size

Category	Baseline Accuracy	Baseline EOD	Fair Model Accuracy	Fair Model EOD	Fairness Improved?
HR	98.56%	0.1667	98.56%	0.0556	Yes (clear improvement)
Accountant	98.92%	0.0714	97.48%	0.0072	Yes (strong improvement)
Business-Development	97.12%	0.6000	95.86%	0.7333	No (much worse)
Information-Technology	97.30%	0.3333	96.76%	0.3333	No change
Fitness	97.12%	0.1250	97.30%	0.1000	Slight improvement

Table 7: Fairness and Accuracy Comparison Across Five Job Categories

Conclusion

This project demonstrated that combining advanced text representation techniques with balanced machine-learning workflows can significantly improve the accuracy and reliability of automated resume classification. The SBERT-based classifier achieved strong performance across multiple job categories despite substantial label noise and class imbalance, highlighting the robustness of semantic embeddings for unstructured text. Fairness evaluation further revealed that while some categories benefited from fairness constraints—showing notable reductions in metrics such as Equalized Odds Difference—others exhibited unstable behavior due to limited sample sizes and uneven subgroup distributions. Together, these findings emphasize that both predictive performance and fairness outcomes are heavily influenced by data quality and representativeness.

The implications of these results extend directly to the broader field of automated hiring systems. As organizations increasingly rely on AI-driven tools for screening and decision-making, ensuring both accuracy and fairness becomes critically important. This work shows that fairness-aware algorithms can meaningfully reduce disparities when applied to sufficiently large and balanced datasets, but they also underscore the need for richer, more consistent data to avoid misleading fairness metrics. Future research should therefore focus on expanding datasets, exploring alternative fairness definitions, and experimenting with more sophisticated debiasing techniques—including adversarial methods, causal inference, and fairness-aware embeddings—to build systems that are not only effective but also equitable and trustworthy. And also could expand fairness evaluation to additional sensitive attributes such as gender or nationality once such data becomes available.

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